

# **Chapter 5 on Instrumentation**

**Unit 5: Connectivity Technology in Instrumentation System  
(6 Hours / 6 Marks)**

# Introduction to Wired and Wireless Communication System

## Wired vs Wireless Communication

| Feature    | Wired Communication                                    | Wireless Communication                                    |
|------------|--------------------------------------------------------|-----------------------------------------------------------|
| Definition | Data transmission using physical cables (copper/fiber) | Data transmission using electromagnetic waves (no cables) |
| Medium     | Twisted pair, coaxial, fiber optic cables              | Radio waves, infrared, Bluetooth, Wi-Fi, etc.             |
| Examples   | Ethernet, HDMI, USB, RS-232                            | Wi-Fi, Bluetooth, Zigbee, 4G/5G, Infrared, LoRa           |

# Comparison

| Factor            | Wired                                           | Wireless                                                     |
|-------------------|-------------------------------------------------|--------------------------------------------------------------|
| Speed             | Generally faster (up to Gbps, especially fiber) | Slower (depends on tech: Wi-Fi < 6 Gbps, Bluetooth < 3 Mbps) |
| Latency           | Lower latency                                   | Higher latency                                               |
| Security          | More secure (physical access required)          | Less secure, easier to intercept unless encrypted            |
| Installation      | Difficult (cable routing, connectors)           | Easier and more flexible                                     |
| Mobility          | Limited (fixed location)                        | High (supports portable/mobile devices)                      |
| Interference      | Minimal (except with electromagnetic noise)     | Prone to interference (from other wireless signals/devices)  |
| Maintenance       | Can be complex (cable wear, breakage)           | Easier (less hardware)                                       |
| Cost (short term) | Higher (cabling, installation labor)            | Lower (fewer physical materials)                             |
| Scalability       | Harder to scale (each new device needs wiring)  | Easier (just connect to network wirelessly)                  |

# Application or Use Case

## **Wired Communication:**

- Data centers
- Industrial control systems
- Security cameras (PoE)
- Desktop PC networks

## **Wireless Communication:**

- Smartphones, tablets
- Smart home devices (IoT)
- Wi-Fi-enabled laptops
- Remote sensing in agriculture

# Twisted Pair Cable

## Definition:

- A cable made of two insulated copper wires twisted together to reduce electromagnetic interference (EMI).

## Types:

- **UTP** – Unshielded Twisted Pair (common in LANs)
- **STP** – Shielded Twisted Pair (has shielding for better protection)

## Characteristics

| Property  | Details                          |
|-----------|----------------------------------|
| Medium    | Copper wires                     |
| Data Rate | Up to 10 Gbps (Cat 6a or higher) |
| Range     | Up to 100 meters                 |
| Cost      | Low                              |
| Use       | Telephones, LANs, CCTV           |

## **Advantages:**

- Inexpensive and easy to install
- Flexible and lightweight
- Good for short-distance data transmission

## **Disadvantages:**

- Susceptible to electromagnetic interference (especially UTP)
- Lower bandwidth and speed compared to optical fiber
- Limited range

# Coaxial Cable

## Definition:

- A cable with a central copper conductor, surrounded by insulation, a metallic shield, and an outer jacket.
- Common in TV and internet connections.

## Characteristics:

### Property

**Medium**

**Data Rate**

**Range**

**Cost**

**Use**

### Details

Copper with shield

Up to 10 Mbps to 1 Gbps

Up to 500 meters

Moderate

Cable TV, broadband internet, CCTV

## **Advantages:**

- Better protection from EMI than twisted pair
- Higher bandwidth and longer range
- Durable for outdoor/underground use

## **Disadvantages:**

- Thicker and less flexible than twisted pair
- Higher cost
- More difficult to install



# Optical Fiber Cable

## Definition:

- A cable that transmits data as pulses of light through glass or plastic fibers, offering ultra-high-speed communication.
- **Types:**
- **Single-mode fiber** – Long-distance, high-speed
- **Multi-mode fiber** – Shorter distance, wider core

## Characteristics:

| Property  | Details                                                       |
|-----------|---------------------------------------------------------------|
| Medium    | Glass or plastic fiber                                        |
| Data Rate | > 100 Gbps (extremely high)                                   |
| Range     | Tens of kilometers                                            |
| Cost      | High (but decreasing over time)                               |
| Use       | Internet backbones, submarine cables, hospitals, data centers |

## **Advantages:**

- Very high bandwidth and speed
- Immune to electromagnetic interference
- Supports long-distance transmission

## **Disadvantages:**

- Expensive equipment and maintenance
- Fragile and requires careful handling
- Complex installation and termination

# Comparison Summary Table

| Feature        | Twisted Pair                 | Coaxial Cable                 | Optical Fiber       |
|----------------|------------------------------|-------------------------------|---------------------|
| Medium         | Copper wires                 | Copper with shielding         | Glass/plastic fiber |
| Speed          | Low to medium                | Medium                        | Very high           |
| EMI Resistance | Low (UTP),<br>Medium (STP)   | High                          | Very High           |
| Distance       | Short ( $\leq 100\text{m}$ ) | Medium ( $\leq 500\text{m}$ ) | Long (km range)     |
| Cost           | Low                          | Moderate                      | High                |
| Installation   | Easy                         | Moderate                      | Difficult/fragile   |
| Use Cases      | LANs, phones                 | TV, internet, CCTV            | Data centers, ISPs  |

# Radio Waves

## Definition:

- Radio waves are **electromagnetic waves** with the **longest wavelengths** (from about 1 millimeter to 100 kilometers) and the **lowest frequencies** (3 Hz to 300 GHz).
- They are commonly used in **communication systems**.

## Characteristics:

- Frequency Range: **3 Hz – 300 GHz**
- Wavelength: **1 mm – 100 km**
- Travel long distances
- Can penetrate through non-metallic materials
- Omnidirectional

## **Advantages:**

- Used for **AM/FM radio, TV, mobile communication**
- Can travel long distances with **minimal attenuation**
- Works in **all weather conditions**

## **Disadvantages:**

- **Lower bandwidth** → lower data transfer rate
- **Prone to interference** from electrical equipment and atmospheric conditions

# Microwaves

## Definition:

- Microwaves are **high-frequency radio waves** with shorter wavelengths, used in **satellite communication, radar, and microwave ovens**.

## Characteristics:

- Frequency Range: **300 MHz – 300 GHz**
- Wavelength: **1 mm – 1 m**
- Travel in **straight lines**
- Cannot easily penetrate buildings
- Requires **line-of-sight** for communication

## Advantages:

- **High data capacity**
- Useful in **satellite, radar, and Wi-Fi communication**
- Suitable for **point-to-point** communication

## Disadvantages:

- Affected by **obstacles** (need clear LOS)
- **Rain, fog**, and other weather conditions reduce signal strength (attenuation)
- **Costlier** infrastructure (dishes, antennas)

# Infrared (IR)

## Definition:

- Infrared waves are **electromagnetic waves** just below visible light in the spectrum. They are used in **remote controls, night vision, and short-range wireless communication**.

## Characteristics:

- Frequency Range: **300 GHz – 430 THz**
- Wavelength: **700 nm – 1 mm**
- Requires **direct line-of-sight**
- **Cannot penetrate walls**
- Can carry large amounts of data over short distances



## **Advantages:**

- **Secure**, short-range communication (e.g., TV remotes)
- **Low interference**
- No license required for most uses

## **Disadvantages:**

- **Limited range**
- **Cannot pass through walls or obstacles**
- Affected by **sunlight or heat sources**

## Summary Table:

| Feature     | Radio Waves           | Microwaves                 | Infrared (IR)           |
|-------------|-----------------------|----------------------------|-------------------------|
| Frequency   | 3 Hz – 300 GHz        | 300 MHz – 300 GHz          | 300 GHz – 430 THz       |
| Wavelength  | 1 mm – 100 km         | 1 mm – 1 m                 | 700 nm – 1 mm           |
| Penetration | Good (walls, objects) | Poor (needs line of sight) | Very poor (short range) |
| Range       | Long range            | Medium to long (LOS)       | Short range             |
| Data Rate   | Low                   | High                       | High (short range only) |
| Common Uses | Radio, TV, mobile     | Wi-Fi, radar, satellite    | Remote control, sensors |

# Serial Data Transmission

- In a serial data transmission, the data are sent one bit at a time over the transmission channel.
- However, since most processors process data in parallel, the transmitter needs to transform incoming parallel data into serial data and the receiver needs to do the opposite.
- In case of serial transmission data is sent in a serial form i.e. bit by bit on a single line.
- Also, the cost of communication hardware is considerable reduced since only a single wire or channel is require for the serial bit transmission.
- Serial data transmission is slow as compared to parallel transmission. Serial data can be sent synchronously or asynchronously.

## Serial Synchronous Data Transmission

- In serial synchronous data transmission, data is transmitted or received based on a clock signal.
- At a specific rate of data transmission, the transmitting device sends a data bit at each clock pulse.
- In order to interpret the data correctly, the receiving device must know the start and end of each data unit.
- The transmitter must know the number of data units to be transferred and the receiver must be synchronized with the data boundaries.
- Therefore, there must be synchronization between the transmitter and receiver. Usually one or more SYNC characters are used to indicate the start of each synchronous data stream or frame of data.
- Transmitter sends a large block of data characters one after the other with no time between characters.
- Transmitting device sends data continuously to the receiving device.
- If the data is not ready to be transmitted, the line is held in marking condition.
- To indicate the start of transmission, the transmitter sends out one or more SYNC characters or a unique bit pattern called a flag, depending on the system being used.
- The receiving device waits for data, when it finds the SYNC characters or the flag then starts interpreting the data which shifts the data following the SYNC characters and converts them to parallel form so they can be read in by a computer.

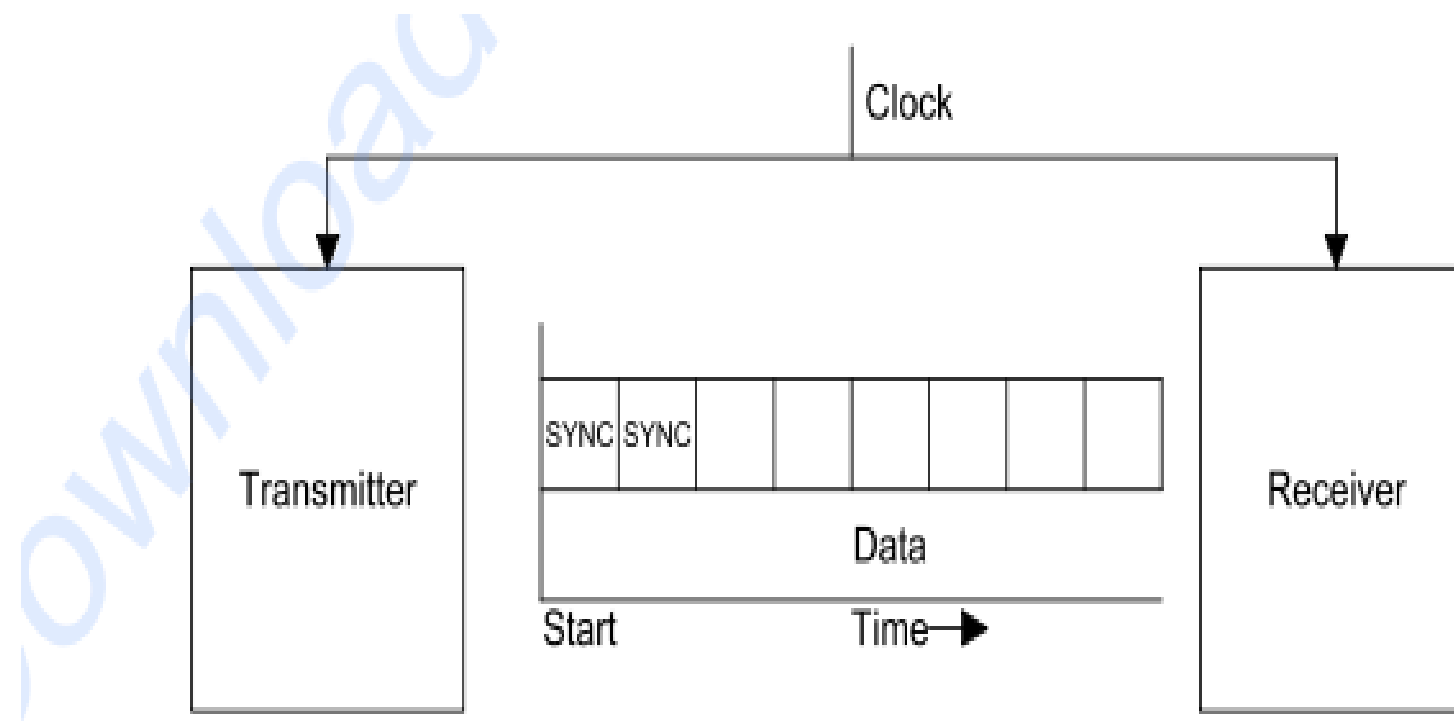


Fig: Synchronous Serial Transmission Format

- Synchronous transmission has the advantage that the timing information is accurately aligned to the received data, allowing operation at much higher data rates.
- It also has the advantage that the receiver tracks any clock drift which may arise (for instance due to temperature variation).
- The penalty is however a more complex interfaces design, and potentially a more difficult interface to configure (since there are many more interface options).
- Data transmission takes place without any gap between two adjacent characters.
- However data is send block by block. A block is a continuous steam of characters or data bit pattern coming at a fixed speed.
- You will find a SYNC bit pattern between any two blocks of data and hence the data transmission is synchronized.
- Synchronous communication is used generally when two computers are communicating to each other at a high speed or a buffered terminal is communicating to the computer.

## **Advantages and Disadvantages of Synchronous Communication**

- Main advantage of Synchronous data communication is the high speed.
- The synchronous communications require high-speed peripherals/devices and a good-quality, high bandwidth communication channel.
- The disadvantage includes the possible in accuracy.
- Because when a receiver goes out of Synchronization, losing tracks of where individual characters begin and end.
- Correction of errors takes additional time.

## Serial Asynchronous Data Transmission

- The receiving device does not need to be synchronized with the transmitting device.
- The transmitting device can send one or more data units when it is ready to send data.
- Each data unit must be formatted i.e. must contain start and stop bits for indicating beginning and the end of data unit.
- It also includes one parity bit to identify odd or even parity of data. To send ASCII character, the framing of data should contain:
  - 1 start bit: Beginning of data
  - 8 bit character: Actual data transferred
  - 1 or 2 stop bits: End of data
- When no data is being sent, the signal line is in a constant high or marking state.
- The beginning of the data character is indicated by the line going low for 1 bit time and this bit is called a start bit.
- The data bits are then sent out on the line one after the other where the least significant bit is sent out first.
- Parity bit should contain to check for errors in received data.
- After the data bit and a parity bit, the signal line is returned high for at least 1 bit time to identify the end of the character, this always high bit is referred to as a stop bit.
- Some older systems use 2 stop bits.



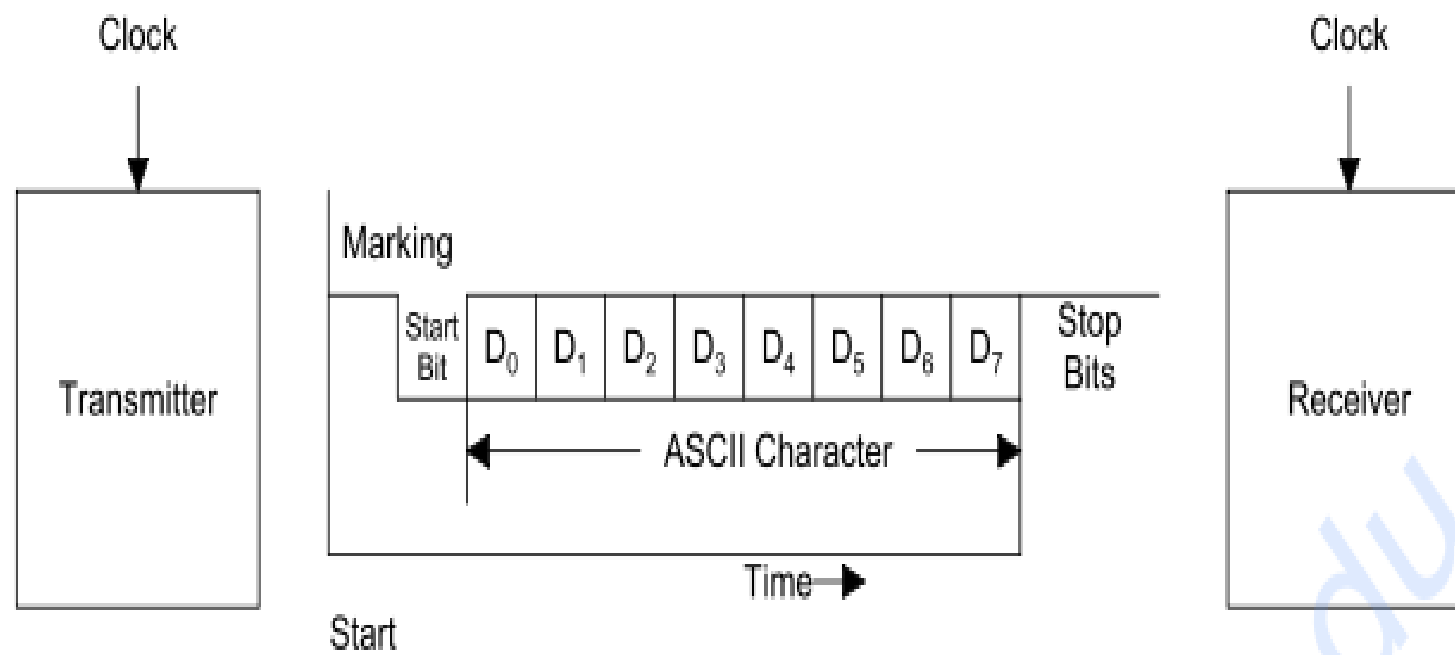


Fig: Asynchronous Serial Transmission Format

- In asynchronous transmission each character is transmitted separately, that is one character at a time.
- The character (8-bits) is preceded by a start bit (1-bit), which tells the receiving end where the character coding begins, and is followed by a stop bit (1 or 2-bits), which tells the receiver where the character coding ends.
- There will be intervals of idle time on the channel shown as gaps.
- Thus there can be gaps between two adjacent characters in the asynchronous communication scheme.
- In this scheme, the bits within the character frame (including start, parity and stop bits) are sent at the baud rate.
- The START bit and STOP bit including gaps allow the receiving and sending computers to synchronize the data transmission.
- Asynchronous communication is used when slow speed peripherals communicate with the computer.
- The main disadvantage of asynchronous communication is slow speed transmission.
- Asynchronous communication however, does not require the complex and costly hardware equipments as is required for synchronous transmission.

### **Synchronous versus Asynchronous serial data transmission**

| <b>S.N.</b> | <b>Parameter</b>    | <b>Asynchronous</b>                               | <b>Synchronous</b>                             |
|-------------|---------------------|---------------------------------------------------|------------------------------------------------|
| 1.          | Fundamental         | Transmission does not based on clock signal       | Transmission based on clock signal             |
| 2.          | Data Format         | One character at a time                           | Group of characters i.e. a block of characters |
| 3.          | Speed               | Low (< 20 kbps)                                   | High (> 20 kbps)                               |
| 4.          | Framing Information | Start and stop bits are sent with each character. | SYNC characters are sent with each character.  |
| 5.          | Implementation      | Hardware / Software                               | Hardware                                       |

# UART: Universal Asynchronous Receiver/Transmitter

## Definition:

- UART is a **hardware communication protocol** used for **serial communication** between devices.
- It transmits and receives data **asynchronously**, meaning **no shared clock** is required between sender and receiver.
- It is commonly used in **embedded systems** like Arduino, STM32, Raspberry Pi, GPS modules, Bluetooth modules, etc.

# Key Features

| Feature                   | Description                                                       |
|---------------------------|-------------------------------------------------------------------|
| Mode                      | Asynchronous (no clock signal shared)                             |
| Data Transfer             | Serial (bit by bit, LSB first)                                    |
| Number of Wires           | Typically, 2 wires – <b>TX</b> (Transmit), <b>RX</b> (Receive)    |
| Data Bits                 | 5 to 9 bits per frame                                             |
| Baud Rate                 | Configurable speed (e.g., 9600, 115200 bps)                       |
| Start/Stop Bits           | Used to frame data and enable synchronization                     |
| Parity Bit (Optional)     | For simple error detection                                        |
| Full Duplex Support       | Transmit and receive simultaneously                               |
| Hardware/Software Support | Can be implemented in both (hardware UART or software-based UART) |

# Working Principle

- **Data Framing:**

The transmitting UART converts **parallel data** (from the CPU or memory) into **serial form** and adds framing bits (start, parity, stop).

- **Transmission:**

Data is sent **bit by bit** through the **TX line** at a specific **baud rate** (speed).

- **Reception:**

The receiving UART detects the **start bit**, then reads the data bits, checks the **parity bit** (if enabled), and stops at the **stop bit**.

- **Deserialization:**

Received serial bits are converted back into **parallel form** and passed to the receiving device.

## UART Frame Format

| Start Bit | Data Bits (5–9) | Parity Bit (optional) | Stop Bit(s) |

### Example (8N1 format)

- **Start bit:** 1 bit (logic 0)
- **Data bits:** 8 bits
- **Parity bit:** None
- **Stop bit:** 1 bit (logic 1)
- **Total:** **10 bits** per data byte

# Advantages of UART

## Advantage

Simple and Inexpensive

No Clock Signal Needed

Full-Duplex Communication

Widely Supported

Configurable

## Limitations

### Limitation

Limited Distance

Fixed Baud Rate

No Multi-Device Support

Basic Error Detection

## Description

Easy to implement in hardware or software

Only 2 wires required (TX and RX)

Simultaneous send and receive

Found in most microcontrollers and embedded systems

Adjustable baud rate, data bits, stop bits, etc.

## Description

Not suitable for long-distance communication

Both devices must agree on same speed

Only supports point-to-point communication

Only parity bit — not very reliable



# SPI, I2C

- Refer to the chapter 4 for details.

# Comparing SPI to I2C and UART

| SPI                                                                               | I <sup>2</sup> C                                                                                                                     | UART                                                                                       |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Can operate at the highest speeds.                                                | Maximum speed is highly dependent upon physical characteristics of the bus (length, number of devices, pull-up strength, and so on). | Requires baud rate to be agreed upon by both devices in advance of starting communication. |
| Is generally easier to work with than I <sup>2</sup> C.                           | Requires only two communication lines.                                                                                               | Effectively uses zero protocol overhead—very simple to implement.                          |
| No pull-up resistors needed.                                                      | Can support communication between devices operating from different voltage rails.                                                    | No predefined master/slave—it is up to you to define the protocol.                         |
| Number of slave devices is limited only by number of available SS pins on master. | Number of slave devices is limited by availability of chips with particular slave addresses.                                         | Cannot easily support multiple slave devices.                                              |
| Has built-in Arduino hardware and software support.                               | Has built-in Arduino hardware and software support.                                                                                  | Has built-in Arduino hardware and software support.                                        |

# CAN (Controller Area Network)

- A robust vehicle bus standard designed to allow microcontrollers and devices to communicate with each other **without a host computer**, primarily used in **automotive** and **industrial** systems.

## 1. Basic Components

- **CAN Transceiver**: Converts digital signals from the CAN controller to differential signals on the CAN bus (and vice versa).
- **CAN Controller**: Built-in on many microcontrollers (e.g., STM32, Arduino with MCP2515).
- **CAN Bus**: The physical medium (typically a **twisted pair cable**) that transmits differential signals.

## **2. Wiring**

- Two Wires:**

- CAN\_H (High)
- CAN\_L (Low)

- Topology:** Bus topology (not star)

- Termination Resistors:**

- 120Ω at each end of the bus to prevent signal reflection.

- Typical Voltage:**

- When idle: Both wires ~2.5V
- During transmission: Differential voltage  
(CAN\_H > CAN\_L for dominant bit)

## **3. Cable Type**

- Twisted Pair Cable**

- Minimizes EMI (electromagnetic interference)
- Common in industrial setups

## 4. Connectors

- No fixed standard, but common:
  - **DB9** (in industrial use)
  - **Screw terminals** (in dev boards)
  - **Custom connectors** in vehicles

## 5. Length and Speed Trade-Off

| Speed (bps) | Max Length  |
|-------------|-------------|
| 1 Mbps      | ~40 meters  |
| 500 kbps    | ~100 meters |
| 125 kbps    | ~500 meters |

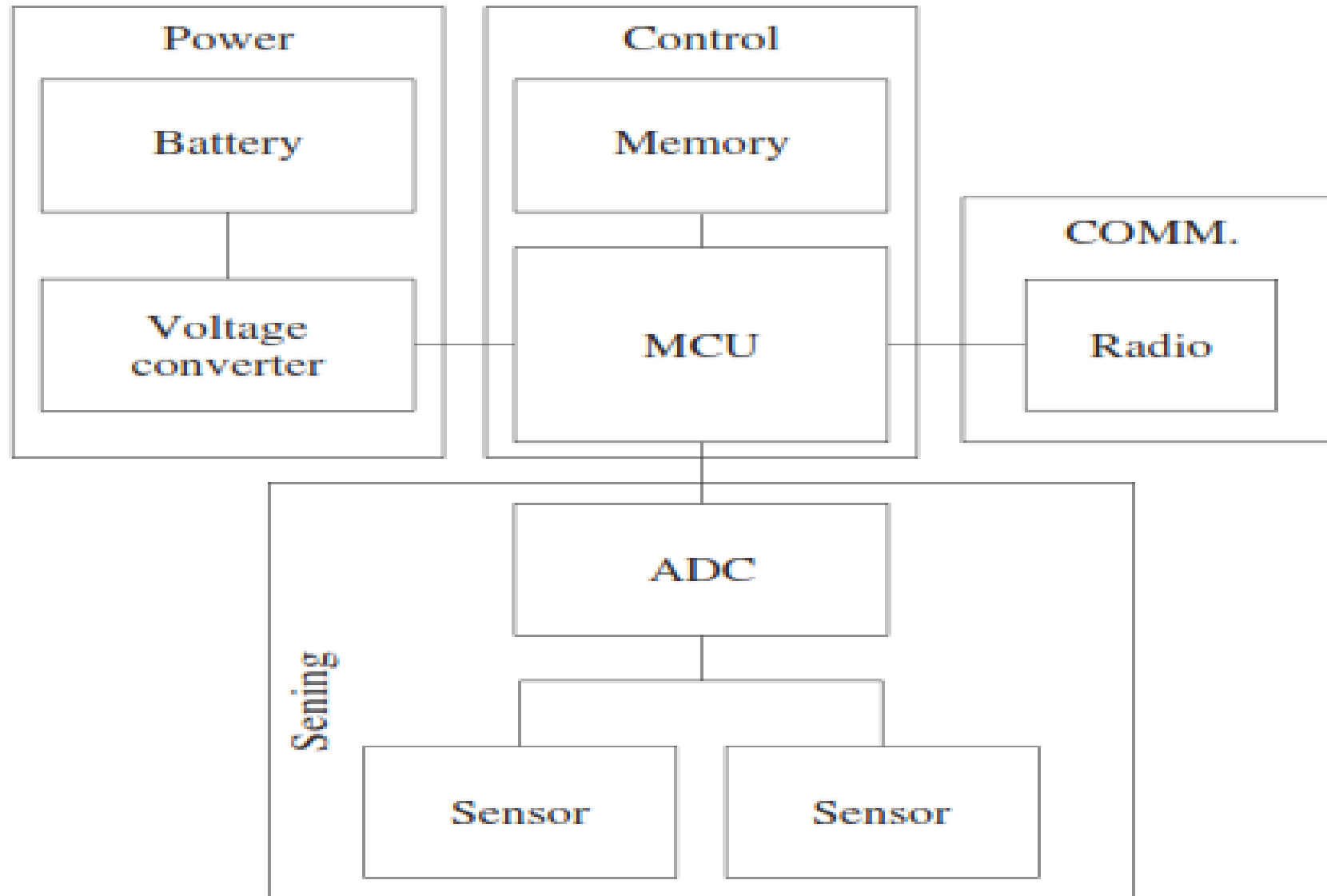
## Advantages of CAN Wired Connectivity

- Real-time, prioritized communication
- High reliability and fault tolerance
- Multi-master bus (any node can initiate)
- Built-in error detection

# WSN (Wireless Sensor Network)

- A network of spatially distributed autonomous sensors that monitor physical or environmental conditions like **temperature, humidity, pressure, motion**, etc., and cooperatively pass their data through the network to a central point (called **sink** or **base station**).
- Wireless sensor networks (WSN), as the name suggests, is a networking paradigm that makes use of spatially distributed sensors for gathering information concerning the immediate environment of the sensors and collecting the information centrally.
- Here, the sensors are not standalone devices but a combination of sensors, processors, and radio units—referred to as sensor nodes—sensing the environment and communicating the sensed data wirelessly to a remote location, which may or may not be connected to a backbone network.

# Typical Block Diagram of WSN



# Key Components

- **Sensor Nodes:** Small, battery-powered devices with:
  - Sensing unit (e.g., temperature, gas, motion)
  - Processing unit (microcontroller)
  - Wireless transceiver (e.g., ZigBee, LoRa, Wi-Fi, BLE)
  - Power unit (usually battery)
- **Base Station (Sink Node):**
  - Collects data from sensor nodes
  - Connects to the Internet or local storage
  - Usually has more resources (e.g., power, computation)
- **Communication Protocols:**
  - ZigBee
  - Bluetooth Low Energy (BLE)
  - LoRa / LoRaWAN
  - Wi-Fi
  - 6LoWPAN



# Types of WSN

| Type        | Description                                     |
|-------------|-------------------------------------------------|
| Terrestrial | Sensors on ground (e.g., agriculture, military) |
| Underground | Buried sensors (e.g., soil moisture, pipelines) |
| Underwater  | For marine exploration (uses acoustic waves)    |
| Mobile      | Moving sensors (e.g., robots, drones)           |
| Multimedia  | Capture video/audio (e.g., smart surveillance)  |

## Applications of WSN

- **Smart Homes:** Motion, temperature, smoke sensors
- **Smart Agriculture:** Soil moisture, sunlight, rainfall
- **Traffic Monitoring:** Vehicle detection, smart parking
- **Industrial IoT:** Machine health, safety monitoring
- **Environmental Monitoring:** Forest fire, pollution
- **Military:** Surveillance, intrusion detection

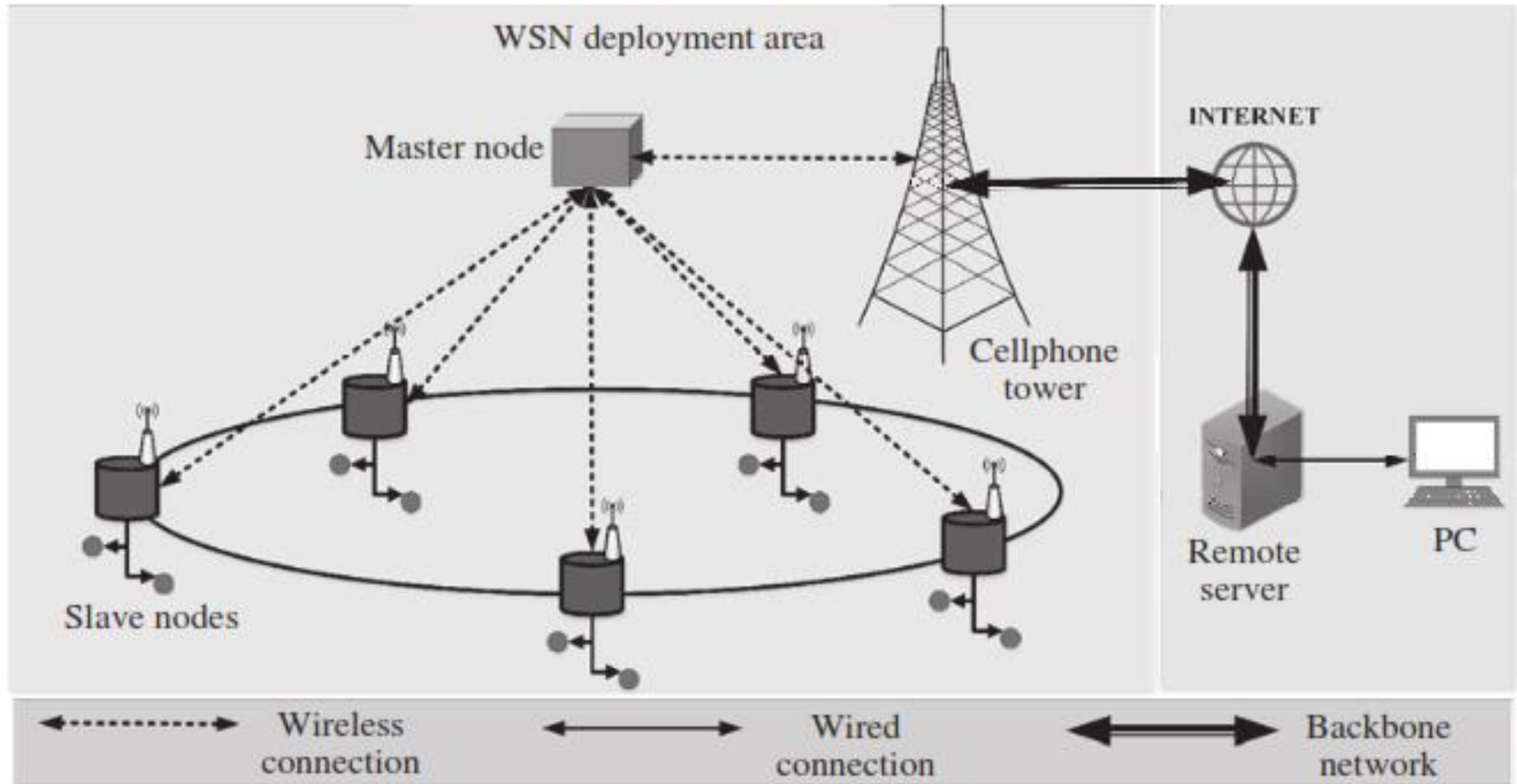
## **Advantages**

- No need for wiring → easier installation
- Scalable (easy to add more nodes)
- Can cover remote or hazardous areas

## **Challenges**

- Limited power (battery life)
- Security vulnerabilities
- Limited bandwidth
- Network reliability in harsh environments

# A Typical WSN Deployment



- Figure shows a typical WSN implementation, where the master node aggregates data from multiple slave nodes, forwards it to a remote server utilizing access to the Internet through cellular connectivity.
- The stored data on the server can be visualized by a user or a subscriber to the system from anywhere in the world over the Internet.
- WSNs mainly follow a system of communication known as master–slave architecture.
- In a master–slave architecture, a single aggregator node, the master, is responsible for collecting data from various sensor nodes under its dominion or range of operations.
- The sensor nodes under the range of the master node are referred to as slave nodes.
- Multiple slave nodes communicate to the master node using low power short-range wireless radios such as Zigbee, Bluetooth, and WiFi for transferring their sensed data to a remote central server.
- Often, in popular WSN architectures, the master node connects the WSN to the Internet and acts as the gateway for the WSN.
- Upon collecting data from the slave nodes, the master node pushes the aggregated data to a remotely located central server using the Internet.
- The master node may be linked to the Internet through cellular connections, another gateway, or directly through a backbone infrastructure.

# **Distinguished Features of WSN**

- **Fault Tolerance**
- **Scalability**
- **Long lifetime**
- **Security**
- **Programmability**
- **Affordability**
- **Heterogeneity**
- **Mobility**

# RF Modem (Radio Frequency Modem)

- An **RF Modem** (Radio Frequency Modulator-Demodulator) is a device that **transmits and receives digital data using radio waves**.
- It's commonly used for **wireless serial communication** between embedded systems, computers, or microcontrollers like Arduino, STM32, etc.
- It converts digital data into **modulated RF signals** for transmission and then demodulates received RF signals back into digital data.

## Typical Use-Case Example

1. Send sensor data from a **remote Arduino** to a **base station**
2. Control a robot wirelessly
3. Communicate between two PCs or microcontrollers wirelessly without Wi-Fi or GSM

## Main Components

- **Microcontroller Interface (UART/USART)** – for digital input/output
- **RF Transmitter/Receiver** – for wireless communication
- **Antenna** – to send/receive signals
- **Power Supply** – usually 3.3V or 5V

## Popular RF Modems

| Modem        | Frequency       | Range        | Data Rate          | Notes                       |
|--------------|-----------------|--------------|--------------------|-----------------------------|
| HC-12        | 433 MHz         | 1 km+ (LOS)  | 1.2–115.2 kbps     | Configurable, UART-based    |
| LoRa (Ra-02) | 433/868/915 MHz | 2–10 km      | ~300 bps–62.5 kbps | Ultra-long range, low power |
| XBee         | 2.4 GHz         | 100 m–1.6 km | Up to 250 kbps     | Zigbee protocol             |
| NRF24L01+    | 2.4 GHz         | 100–1000 m   | Up to 2 Mbps       | SPI interface, low cost     |

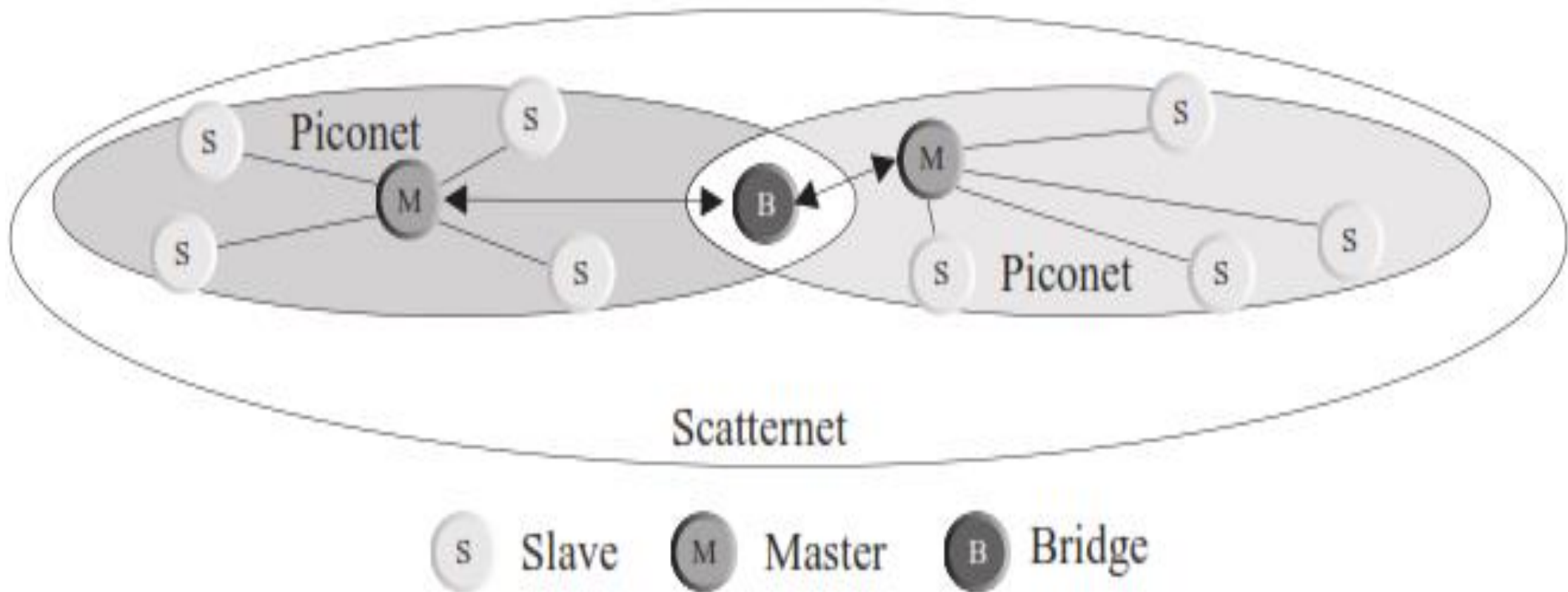
# Bluetooth

- Bluetooth is defined by the IEEE 802.15.1 standard and is a short-range wireless communication technology operating at low power to enable communication among two or more Bluetooth-enabled devices.
- It was initially developed as a cable replacement technology for data communication between two or more mobile devices such as smartphones and laptops.
- This standard allows the transmission of data as well as voice-over short distances.
- Bluetooth functions on the 2.4 GHz ISM band and has a range of approximately 10 m.
- The transmission of data is done through frequency hopping spread spectrum (FHSS), which also reduces the interference caused by other devices functioning in the 2.4 GHz band.



- The data is divided into packets before transmitting them by Bluetooth.
- The packets are transmitted over the 79 designated channels, each 1MHz wide in the 2.4 GHz band.
- Adaptive frequency hopping (AFH) enables this standard to perform 800 hops per second over these channels.
- Initial versions of this standard followed Gaussian frequency shift keying (GFSK) modulation, which was known as the basic rate (BR) mode, and was capable of data rates of up to 1 Mbps.
- However, with the development of newer variants, modulation schemes such as  $\pi/4$  DQPSK (differential quadrature phase shift keying) and 8-DPSK (differential phase shift keying) were adopted, which enabled data rates of 2 Mbps and 3 Mbps respectively.

# The Bluetooth Device Network Architecture

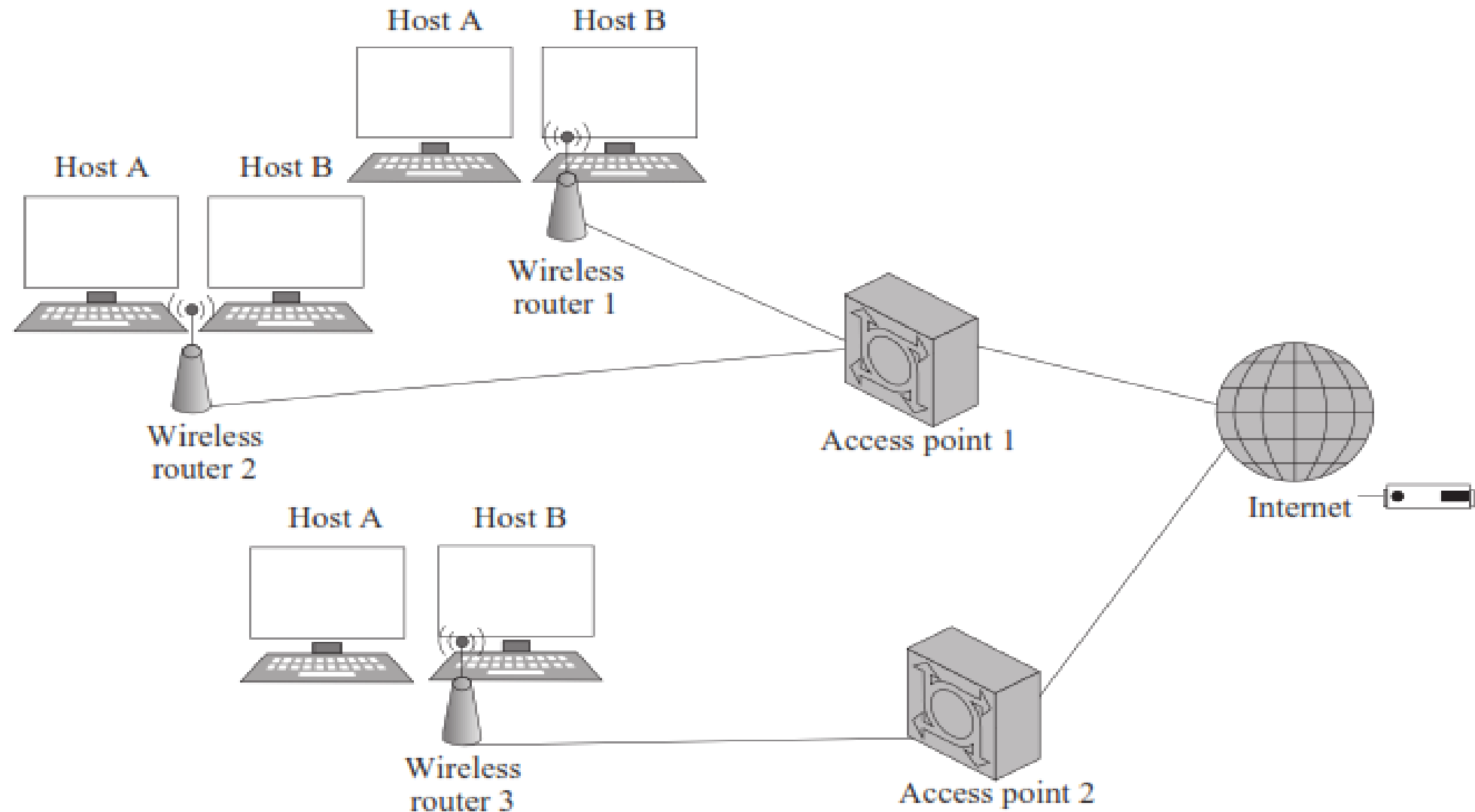


- Bluetooth follows a **master–slave** architecture (shown in figure).
- It enables a small network, which can accommodate seven slave devices simultaneously with a single master node.
- A slave node in one **piconet** cannot be part of another piconet at the same time, that is, it can have a single master node at a time.
- This network is known as a personal area network (PAN) or piconet.
- All the devices in a piconet share the master node's clock.
- Two piconets can be joined using a bridge.
- The whole network is also referred to as a scatternet.
- Bluetooth Low Energy (**BLE**), the advanced variant of Bluetooth has 2 MHz wide bands, which can accommodate 40 channels.
- Its features include low energy consumption, low cost, multivendor interoperability, and an enhanced range of operations.
- Bluetooth connections are encrypted and prevent eavesdropping of communications between devices.
- The inclusion of service-level security adds an additional layer of security by restricting the usage and device features and activities.

# Wi-Fi

- Wi-Fi or WiFi is technically referred to by its standard, IEEE 802.11, and is a wireless technology for wireless local area networking of nodes and devices built upon similar standards.
- Wi-Fi utilizes the 2.4 GHz ultra high frequency (UHF) band or the 5.8 GHz super high frequency (SHF) ISM radio bands for communication.
- For operation, these bands in Wi-Fi are subdivided into multiple channels.
- The communication over each of these channels is achieved by multiple devices simultaneously using time-sharing based TDMA multiplexing.
- It uses CSMA/CA for channel access.
- Various versions of IEEE 802.11 have been popularly adapted, such as a/b/g/n.
- The IEEE 802.11a achieves a data rate of 54 Mbps and works on the 5 GHz band using OFDM for communication.
- IEEE 802.11b achieves a data rate of 11 Mbps and operates on the 2.4 GHz band.
- Similarly, IEEE 802.11g also works on the 2.4 GHz band but achieves higher data rates of 54 Mbps using OFDM.
- Finally, the newest version, IEEE 802.11n, can transmit data at a rate of 140 Mbps on the 5 GHz band.

# The Wi-Fi Deployment Architecture

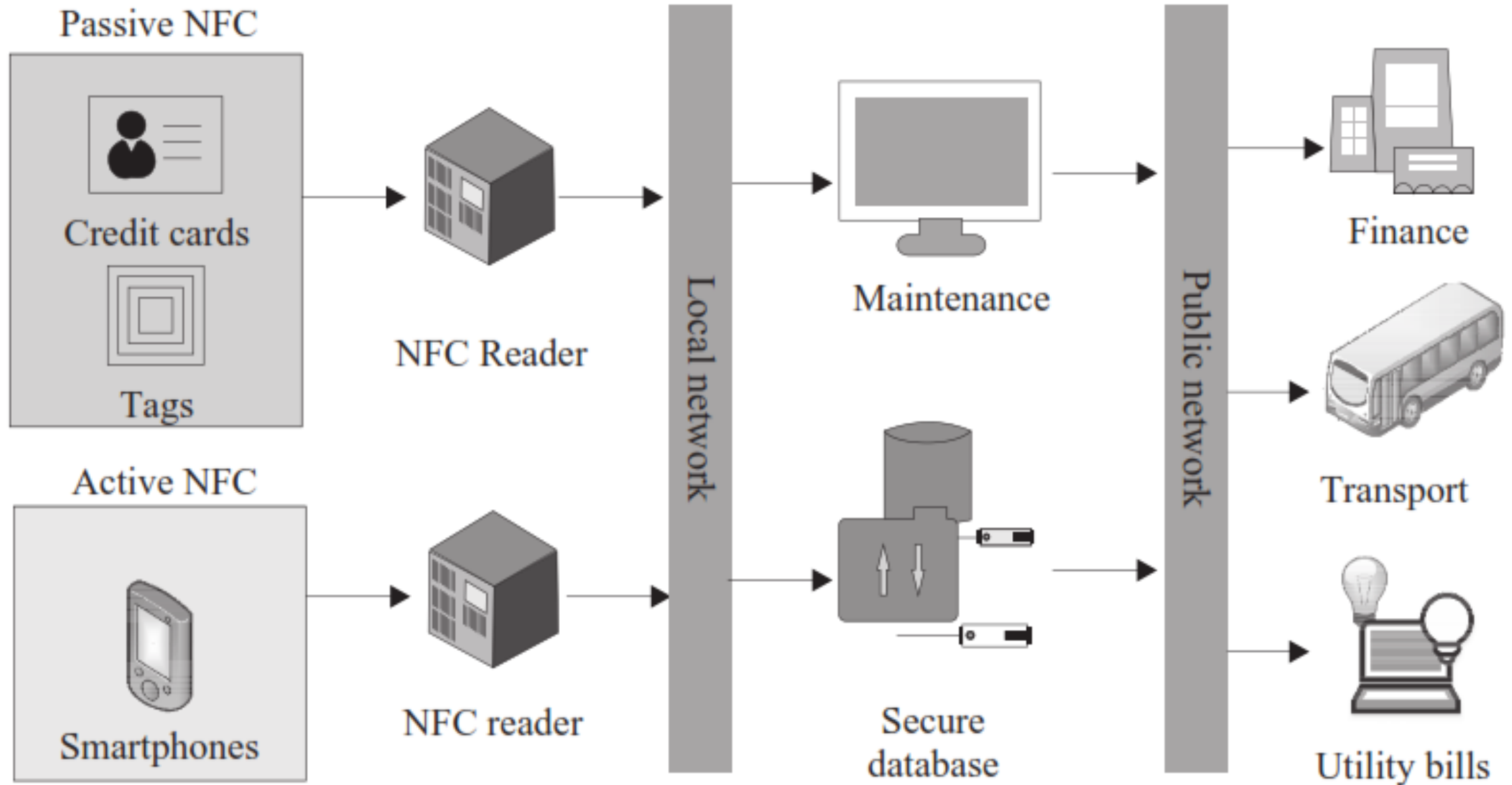


- Wi-Fi devices can network using a technology referred to as wireless LAN (WLAN), as shown in Figure.
- A Wi-Fi enabled device has to connect to a wireless access point, which connects the device to the WLAN.
- WLAN is then responsible for forwarding the messages from the devices to and fro between the devices and the Internet.

# NFC (Near Field Communication )

- NFC was jointly developed by Philips and Sony as a short-range wireless connectivity standard, enabling peer-to-peer (P2P) data exchange network.
- Communication between NFC devices is achieved by the principle of magnetic induction, whenever the devices are brought close to one another.
- NFC can also be used with other wireless technologies such as Wi-Fi after establishing and configuring the P2P network.
- The communication between compatible devices requires a pair of transmitting and receiving devices.
- The typical NFC operating frequency for data is 13.56 MHz, which supports data rates of 106, 212, or 424 kbps.
- NFC devices can be grouped into two types: 1) passive NFC and 2) active NFC.
- Figure shows the various NFC types, components, and its usage.
- A small electric current is emitted by the NFC reader, which creates a magnetic field that acts as a bridge in the physical space between two NFC devices.
- The generated EM (electromagnetic) field is converted back into electrical impulses through another coil on the client device. Data such as identifiers, messages, currency, status, and others can be transmitted using NFCs.
- NFC communication and pairing are speedy due to the use of inductive coupling and the absence of manual pairing.

# An outline of the NFC operation and communication



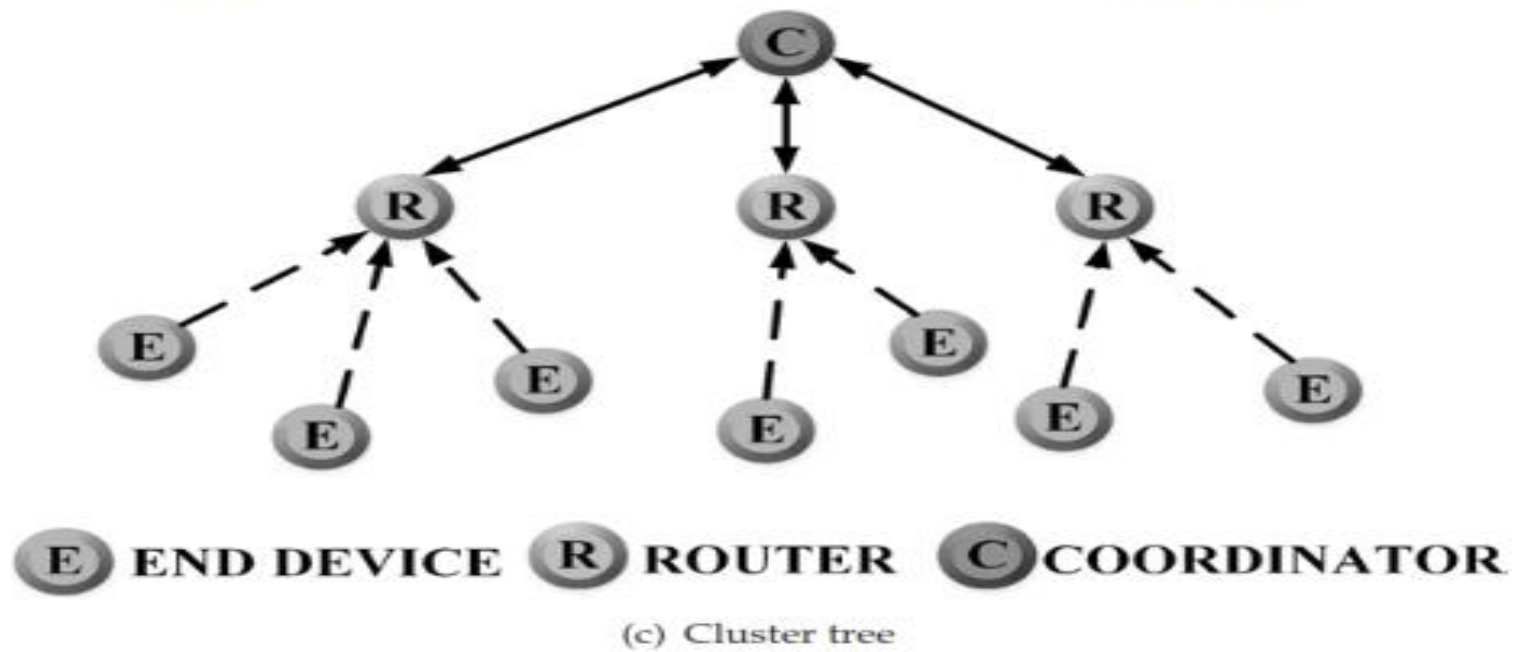
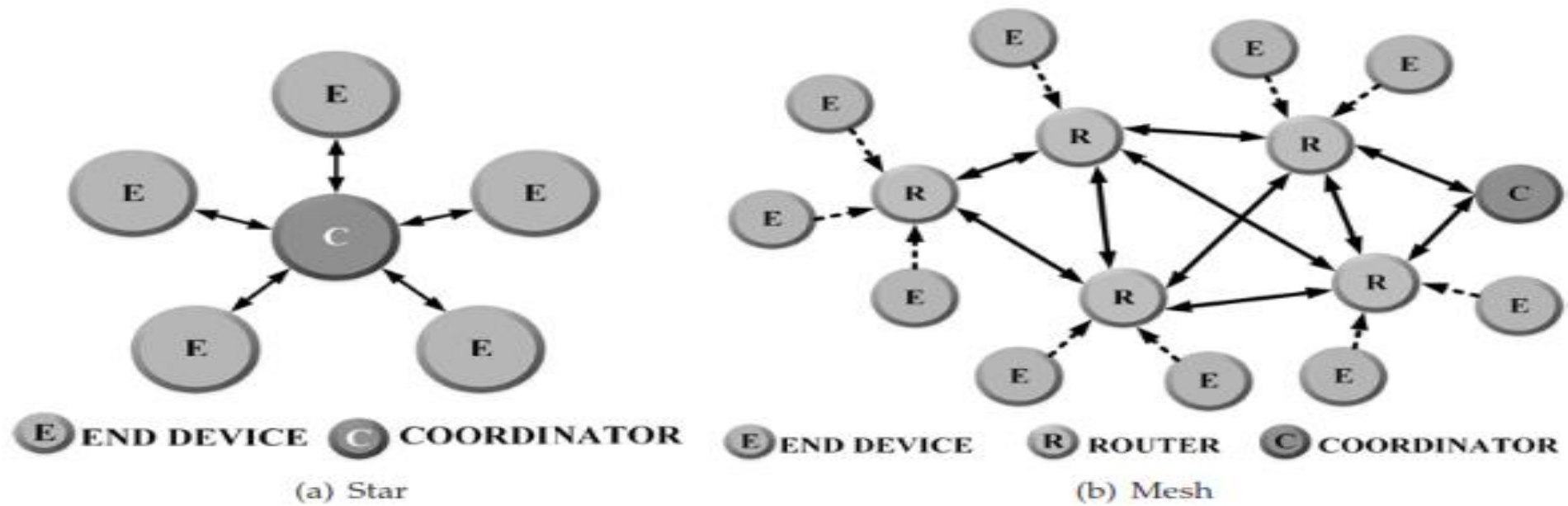


- Passive NFC devices do not need a power source for communicating with the NFC reader.
- Tags and other small transmitters can act as passive NFC devices.
- However, passive devices cannot process information; they simply store information, which is read by an NFC reader.
- In contrast, active NFC devices can communicate with active as well as passive NFC devices.
- Active devices are capable of reading as well as writing data to other NFC terminals or devices.
- Some of the most commonly used NFC platforms are smartphones, public transport card readers, and commercial touch payment terminals.
- NFC currently supports three information exchange modes: 1) **peer-to-peer**, 2) **read/write**, and 3) **card emulation**.
- The **peer-to-peer** mode is commonly used in NFC modes; it enables two NFC devices to exchange information. In the **peer-to-peer** mode of information exchange, the transmitting device goes active while the receiving device becomes passive. During the reverse transfer, both devices change roles.
- The **read/write** mode of information exchange allows only one-way data transmission.
- An active NFC device connects to a passive device to read information from it.
- Finally, the **card emulation** mode enables an NFC device (generally, smartphones) to act as a contactless credit card and make payments using just a simple tap on an NFC reader.

# Zigbee

- The Zigbee radio communication is designed for enabling wireless personal area networks (WPANs).
- It uses the IEEE 802.15.4 standard for defining its physical and medium access control (layers 1 and 2 of the OSI stack).
- Zigbee finds common usage in sensor and control networks.
- It was designed for low-powered mesh networks at low cost, which can be broadly implemented for controlling and monitoring applications, typically in the range of 10–100 meters.
- The PHY and MAC layers in this communication are designed to handle multiple low data rate operating devices.
- The frequencies of 2.4 GHz, 902–928 MHz or 868 MHz are commonly associated with Zigbee WPAN operations.
- The Zigbee commonly uses 250 kbps data rate which is optimal for both periodic and intermittent full-duplex data transmission between two Zigbee entities.

- Zigbee supports various network configurations such as master-to-master communication or master-to-slave communication.
- Several network topologies are supported in Zigbee, namely the star (Figure (a)), mesh (Figure (b)), and cluster tree (Figure (c)).
- Any of the supported topologies may consist of a single or multiple coordinators.
- In star topology, a coordinator initiates and manages the other devices in the Zigbee network.
- The other devices which communicate with the coordinator are called end devices.
- As the star topology is easy to maintain and deploy, it finds widespread usage in applications where a single central controller manages multiple devices.

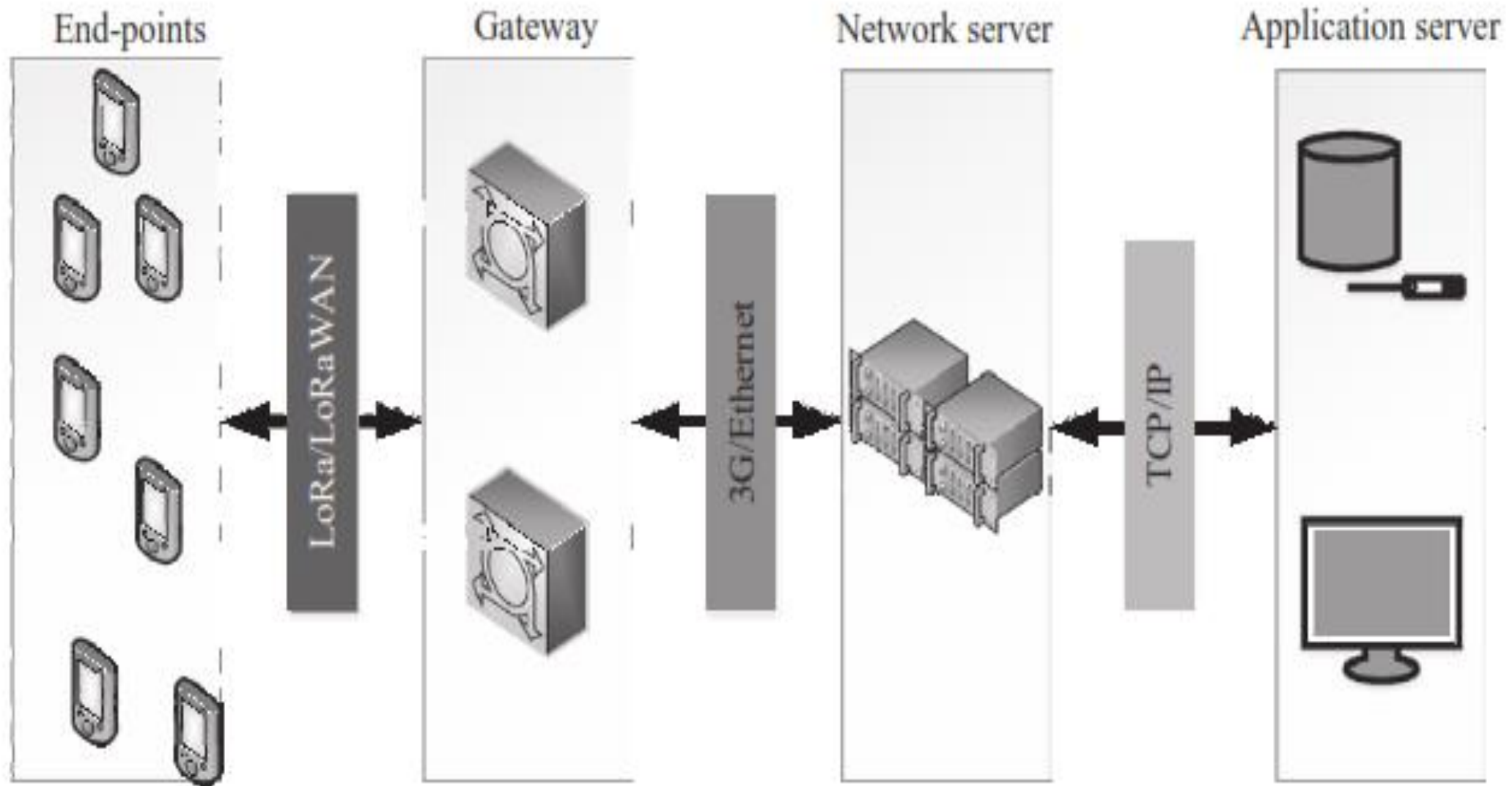


# LoRa (Long Range)

- LoRa is a patented wireless technology for communication developed by Cycleo of Grenoble, France for cellular-type communications aimed at providing connectivity to M2M and IoT solutions.
- It is a sub-GHz wireless technology that operationally uses the 169 MHz, 433 MHz, 868 MHz, and 915 MHz frequency bands for communication.
- LoRa uses bi-directional communication links symmetrically and a spread spectrum with a 125 kHz wideband for operating.
- Applications such as electric grid monitoring are typically suited for utilizing LoRa for communications.
- Typical communication of LoRa devices ranges from 15 to 20 km, with support for millions of devices.

- It is a spread spectrum technology with a broader band (usually 125 kHz or more).
- LoRa achieves high receiver sensitivity by utilizing frequency-modulated chirp coding gain.
- LoRa devices provide excellent support for mobility, which makes them very useful for applications such as asset tracking and asset management.
- In comparison with similar technologies such as NB-IoT, LoRa devices have significantly higher battery lives, but these devices have low data rates (27 to 50 kbps) and longer latency times.

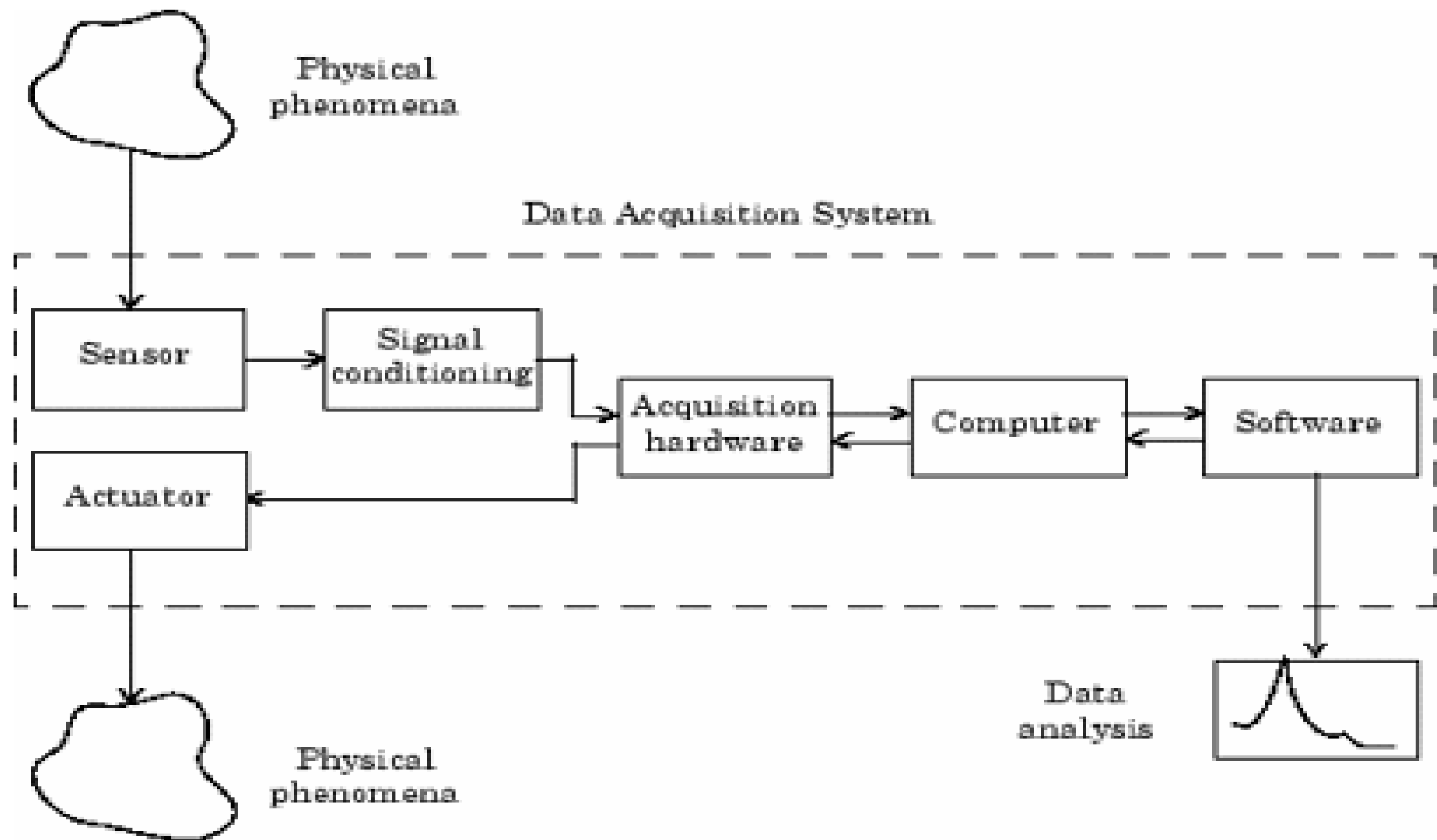
# A Typical LoRa Deployment and Communication Architecture



# Data Acquisition System

- Process of sampling signals that measure real world physical conditions and converting the samples into digital numeric values that can be manipulated by a computer.
- A data acquisition system consists of many components that are integrated to:
  - Sense physical variables (use of transducers)
  - Condition the electrical signal to make it readable by an A/D board
  - Convert the signal into a digital format acceptable by a computer
  - Process, analyse, store, and display the acquired data with the help of software





## Data Acquisition Hardware

- At the heart of any data acquisition system lies the data acquisition hardware.
- The main function of this hardware is to convert analog signals to digital signals, and to convert digital signals to analog signals.

## Sensors and Actuators (transducers)

- Sensors and actuators can both be *transducers*. A *transducer* is a *device that* converts input energy of one form into output energy of another form.
- For example, a microphone is a sensor that converts sound energy (in the form of pressure) into electrical energy, while a loudspeaker is an actuator that converts electrical energy into sound energy.

## Computer

- The computer provides a processor, a system clock, a bus to transfer data, and memory and disk space to store data.

## **Signal Conditioning Hardware**

- Sensor signals are often incompatible with data acquisition hardware.
- To overcome this incompatibility, the signal must be conditioned.
- For example, you might need to condition an input signal by amplifying it or by removing unwanted frequency components.
- Output signals might need conditioning as well.
- However, only input signals conditioning is discussed in this chapter.

## **Software**

- Data acquisition software allows you to exchange information between the computer and the hardware.
- For example, typical software allows you to configure the sampling rate of your board, and acquire a predefined amount of data.

# Data Loggers

- Data logger automatically makes a record of the readings of instruments located at different parts of plant.
- Data logger measures and record data effortlessly as quickly, as often, and as accurately as desired.
- These devices measure electrical output from transducer, give plant performance computation, logic analysis of alarm conditions, passes information (reading) to computer for further processing etc.
- So they are used in power generation plant, petro-chemical installations, real time processing plants etc.

## **Characteristics of Data Logger**

- Modularity
- Reliability and Raggedness
- Accuracy
- Management Tool
- Easy to Use

## **Application of Data Logger**

- Weather station recording e.g. wind speed, wind direction, temperature, relative humidity
- Hydrographical recording e.g. water level, depth, water flow PH, conductivity
- Soil moisture level
- Gas pressure
- Environmental Monitoring

# Basic Operation of Data Logger

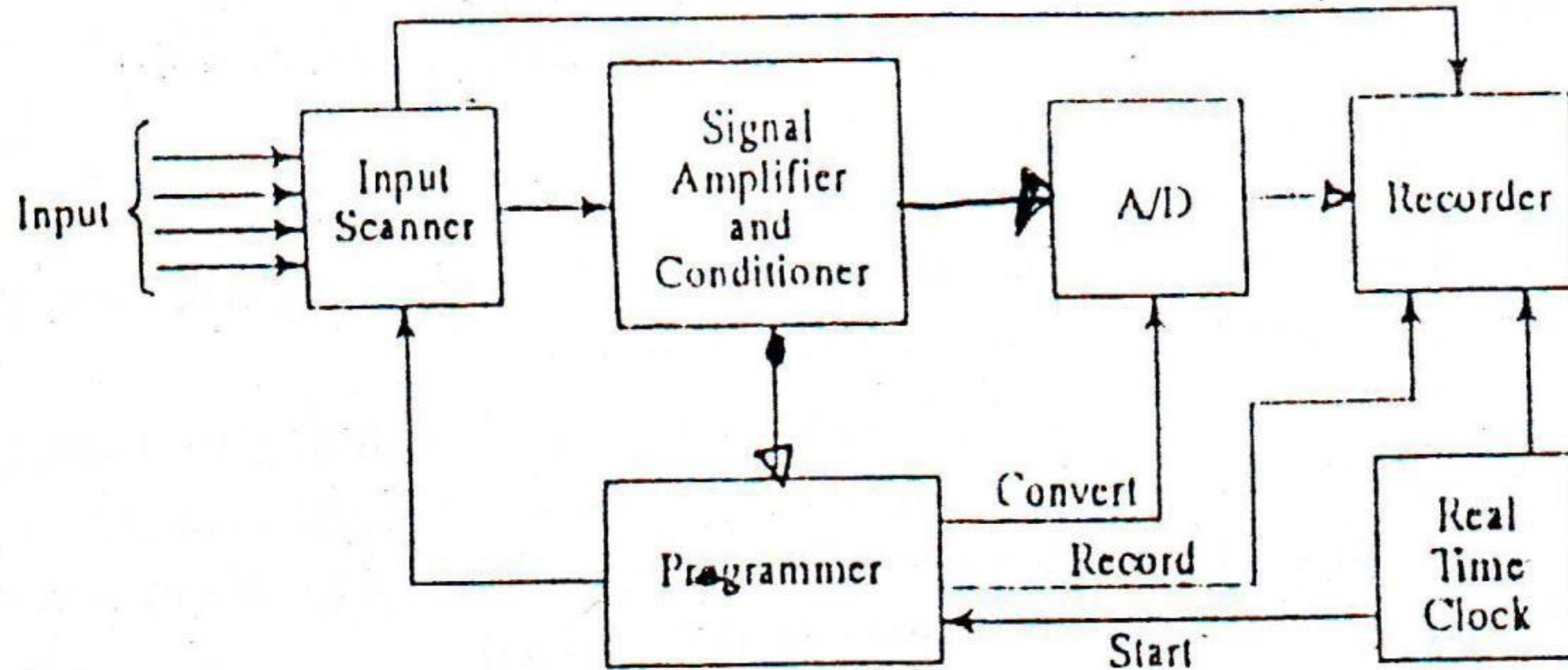


Fig. 17.27 Block Diagram of a Data Logger

## **Input Signals**

- May be Pressure, transducers, Thermocouple AC signal, Signals from relay, switch, Tachometer pulses etc.

## **Input Scanner**

- It is an automatic sequence switch which selects each signal in turn. Modern scanner have input scanner which can scan at a rate of 150 inputs per second.
- Characteristics of input scanner may be:
  - Low closed resistance
  - High open circuit resistance
  - Low contact potential
  - Negligible interaction between switch, enter going signal and input signal
  - Short operating time
  - Negligible contact bounce
  - Long operation life

## Signal Amplifier & Conditioner

- Amplifier for gain adjustment i.e. low level signal amplified up to 5v output.
- Characteristics are:
  - Precise and stable DC gain
  - High SNR
  - High CMMR
  - Low DC drift
  - Low output impedance
  - High input impedance
  - Good linearity
  - Wide bandwidth
- Conditioner for scaling linear transducer or correcting curvature of non linear transducer i.e. signal is changed to more linear form and suitable for digital analysis. Characteristics are:
  - Linear scale
  - Correcting the curvature of non linear transducer
  - It may include sample and hold circuit



## **A/D Converter**

- Converts analog sample into digital data. Characteristics are:
  - Resolution
  - Accuracy
  - Conversion time
  - Full scale output voltage
  - Linearity

## **Recorder**

- Output from data logger may be recorded in any of following:
  - Typewriter, strip printer, digital tape recorder, punched tape, computer (hard drive), magnetic tape etc.
- Characteristics are:
  - Speed
  - Memory
  - Writing technique (Serial / Parallel)

# Programmer

- Control all units of data conversion and operation
- Microcontroller or microprocessor based system
- Basic units: main frames, front panel assembly, power supply unit, scanner controller, input interface etc.
- Operation performed by programmer:
  - Set amplifier
  - Set linearity factor
  - Set high and low alarm value
  - Start A/D conversion
  - Record reading channel
  - Identify channel and time of recording
  - Display recording
  - Reset logger

# Data Compression

- Process of encoding information using fewer bits than an un-encoded representation would use, through specific encoding schemes.
- Reduce consumption of expensive resources such as hard drive and transmission bandwidth.
- Trade-off between compression speed, compressed data size and quality (loss).

- Two types:

## Lossy

- For the case if loss of fidelity is acceptable, e.g.  $6.666666 = 7$
- Examples: Pictures (JPEG), Video (MPEG), Audio (MP3) etc.

## Lossless

- Exploit statistical redundancy in such a way to represent data without error, e.g.  $6.666666 = 6[6]6$
- Examples: zip, rar, Picture (PNG, TIFF), Video (Huff, YUV, AVI) etc.